Highly symmetric hypertopes: distinguished groups and finite constructions

Human Compression project

Abstract

We assemble the general and finite results of Fernandes–Leemans–Weiss in their fixed accepted manuscript. Residue stabilizers supply the C-group and C^+-group assertions; an explicit chamber reflection supplies the converse under its geometric hypotheses. Complete certificates identify all twenty-three finite table entries and the three examples. The synthesis retains the classical string and toroidal classification inputs, corrects the finite captions and one group name, and separates the source's open questions. The two complete component manuscripts and their certificates are part of this proof, rather than replaced by the present summary.

1 The principal statements and their hypotheses

A hypertope is a thin, residually connected incidence geometry of finite rank. Every flag extends to a chamber. Automorphisms in this paper preserve types; correlations that permute types are not included in Aut_I. A regular hypertope has a chamber-transitive automorphism group. A chiral hypertope has exactly two chamber orbits, with adjacent chambers in different orbits. Propagation through residues makes both actions free.

Write $I = \{0, \dots, r-1\}$. For involutions ρ_i put $G_S = \langle \rho_i : i \in S \rangle$. A C-group satisfies $G_S \cap G_T = G_{S \cap T}$ for every $S, T \subseteq I$. For generators $\alpha_1, \dots, \alpha_{r-1}$ put $\alpha_0 = 1$ and

$$H_S = \langle \alpha_i^{-1} \alpha_j : i, j \in S \rangle.$$

The C^+ condition is $H_S \cap H_T = H_{S \cap T}$ for $|S|, |T| \geq 2$, with trivial groups on singleton and empty supports.

Theorem 1. *The principal results in the fixed source hold with the following precise conventions.*

1. *A regular hypertope has distinguished nonidentity involutions ρ_i forming a C-group. Conversely, the standard parabolic coset system of a C-group is a regular hypertope if the designated group is flag-transitive. Its full type-preserving automorphism group is then that group.*

2. *A chiral hypertope of rank at least three has independent distinguished generators α_i forming a C^+-group. Its group is transitive on flags of every proper type, so every proper truncation is flag-transitive; thinness of a truncation is not asserted.*

3. *Let a C^+-group G define the standard coset system with $P_i = H_{I \setminus \{i\}}$. Assume that this system is a hypertope and that G has exactly two chamber orbits. It is chiral, with full group G, if and only if no automorphism of G simultaneously inverts all the α_i. Otherwise its full group is $G \rtimes \langle F \rangle$, where F is a geometric involution interchanging the chamber orbits. If θ is the inverting group automorphism, $FgF^{-1} = \theta(g)$.*

4. *At each of the twelve parameters of source Table 1 and eleven parameters of source Table 2, the specified presentations give the finite regular or chiral objects and full group orders in Table 1 below, with the corrected group identification stated there.*

Proof. We acquire the complete general proof [2] and finite proof [3], together with their distinct reviews. The essential interfaces are as follows. Fix a chamber $C = \{c_i : i \in I\}$ and let $F_{I\setminus S}$ be its complementary flag. With left actions, $C^{ij} = (C^i)^j$, the general proof establishes the identities

$$G_S = \mathrm{Stab}_G(F_{I\setminus S}), \qquad H_S = \mathrm{Stab}_G(F_{I\setminus S})$$

in the regular and chiral cases respectively. In the second identity, $\alpha_i C = C^{0i}$ and $\alpha_i^{-1}\alpha_j C = C^{ij}$. Actual paths within the residue prove the first identity, and even paths prove the second; neither proof assumes an unidentified residue group. Intersections of the stabilizers give both intersection conditions and the independence assertion. The standard C-group converse uses flag transitivity to identify every flag and its residue, and then the original intersection property proves thinness and residue connectivity.

For the third assertion, the same proof identifies *every* chamber as $(g,+) = gC$ or $(g,-) = gC^0$, with

$$(g,+) \sim_i (g\alpha_i^{-1},-).$$

The acquired two-orbit theorem gives the reflection $(g,\pm) \mapsto (\theta(g),\mp)$. Reconstruction from colored chamber adjacencies makes it an incidence automorphism. This is a geometric involution even if θ is the identity. An abstract semidirect product, without this reconstruction, would not pay the converse.

The finite component checks the hypotheses just used for all twenty-three presentations: their exact finite groups, all marked intersections, actual flag completions and residue graphs, and all-relator inversion tests. Its regular completions and named-group maps are explicit isomorphisms. For Table 1 its forty-eight subgroup-product inclusions additionally prove that incidence ordered by type is transitive, giving the full polytope axioms. Its computations use right cosets; the bijection $P_i g \mapsto g^{-1}P_i$ preserves types and incidence and converts them to the left-coset convention of the general proof. Thus these are the same geometries. No new finite search is required for this join. □

2 The finite rows and source corrections

The source's printed g is the order of the full type-preserving group. It is the number of chambers in a regular row, and *half* that number in a chiral row. Thus both captions require the same correction.

p	(a,b)	Table 1 group order	Table 2 group order	status
3	(2,0)	240	120	regular
3	(3,0)	1296	648	regular
3	(4,0)	15360	7680	regular
3	(1,2)	336	168	chiral
3	(1,3)	2184	1092	chiral
3	(1,4)	8064	4032	chiral
3	(2,2)	2880	1440	regular
3	(2,3)	6840	3420	chiral
4	(1,1)	288	—	regular
4	(2,0)	768	384	regular
4	(1,2)	2016	1008	chiral
5	(2,0)	28800	14400	regular

Table 1. Exact positive finite roster; no completeness claim for chiral rows.

The Table 2 group at $(3,2,2)$ is

$$D = \{(s,t) \in S_5 \times S_4 : \operatorname{sgn}(s) = \operatorname{sgn}(t)\},$$

not $A_5 \times S_4$. The finite proof gives a bijective map to D and distinguishes the two groups by the centralizers of their characteristic perfect residuals, of orders 12 and 24. It identifies the groups abbreviated $[1\,1\,2]^3$ and $[1\,1\,2]^4$ by explicit $G(3,3,4)$ and G_{29} models. Every other printed group name is likewise identified by an actual map, including the actions in the split extensions. Each of the eleven Table 2 full groups embeds with index two in its corresponding Table 1 group; the complete marked halving maps and complements are retained.

The three examples are also acquired in full. Example 4.4 has group $(C_3 \times C_3) \rtimes C_2$ and incidence graph $K_{3,3,3}$, with 27 chambers and full group S_3^3. Its designated order-18 group is not flag-transitive and its panels have three completions. Example 4.5 has the printed C-marking of A_6, 720 chambers, 360 panels with four completions and 36 disconnected residues. Example 6.3 has order 72, satisfies the involutory intersection property, and has rotation group equal to the whole group; two rotation subgroups intersect in order 8 instead of the required 4. These are separate reasons why the relevant geometric or index-two hypotheses cannot be discarded.

One additional source assertion about Example 4.5 needs qualification.

Proposition 2. *The rotation marking of Example 4.5 is not a C^+-marking. Consequently the sentence on source page 17 does not supply its claimed C^+ example.*

Proof. Retain the source involutions $\rho_0, \rho_1, \rho_2, \rho_3$ and set $\alpha_i = \rho_0\rho_i$. The acquired finite certificate gives $G = A_6$, the two parabolics $G_{\{0,2,3\}}$, $G_{\{0,1,3\}}$ of order 60, and their intersection $G_{\{0,3\}}$ of order 10. Every index-six subgroup of A_6 is isomorphic to A_5: the action on its six cosets is faithful by simplicity, has image A_6 in S_6, and identifies it with a point stabilizer. For a group generated by involutions, the subgroup generated by their pairwise products is normal and has index at most two. Thus simplicity of A_5 gives

$$H_{\{0,2,3\}} = G_{\{0,2,3\}}, \qquad H_{\{0,1,3\}} = G_{\{0,1,3\}}.$$

Their intersection has order 10, whereas $H_{\{0,3\}} = \langle \rho_0\rho_3 \rangle$ has order 5. Equivalently ρ_0 lies in the former intersection and not the latter. This violates IC^+. The order and group-identification data are already part of Example 4.5's accepted certificate; there is no additional census. □

This corrects the later illustration, not the local group and incidence statements of Example 4.5. In particular it neither proves nor refutes a general converse for arbitrary C^+-groups. Theorem 1(3) keeps its hypertope and exactly-two-orbit hypotheses throughout.

3 Complete source routing and retained inputs

The fixed source is the twenty-two-page accepted author manuscript of [1]. The complete route register accompanies this paper. The nineteen named general clauses are routed as follows:

Source clauses	Complete acquired route
2.1–2.2	Propagation within every residue and flag extension.
3.1–3.5	Cosets and common-core quotient; actual chamber and panel stabilizers; Dehon and Buekenhout–Hermand criteria.
4.1–4.3, 4.6	Regular residue identity, Tits' three-coset criterion, rank-three panel completions and the flag-transitive converse.
5.1–5.2	Retained marked string correspondence and the proved rank-two residue interpretation of a string diagram.
6.1–6.2	Independence for rotations and the even-word kernel at index two.
7.1–7.2	Even residue paths, full support intersections and independence.
8.1–8.2	Proper-flag transitivity and exact two-orbit coordinates.

Construction 8.1 defines an incidence system; flag extension and hypertope properties are not automatic. A coset action is faithful only after factoring its common core. In a nonfaithful chamber-transitive hypertope action the Borel subgroup equals that core, rather than necessarily being trivial. The proof of source Proposition 2.1 uses connectedness of each residue in its induction, not connectedness of the whole incidence graph alone. These qualifications are explicit in the general component.

Four previously established inputs remain attributed premises, rather than new classification proofs: the marked string C-group correspondence (Aschbacher, Schulte and McMullen–Schulte); the rank-three toroidal chirality criterion of Breda D'Azevedo–Nedela; the toroidal hypermap families of Corn–Singerman; and McMullen–Schulte's finite regular locally toroidal polytope classification at $p = 3, 4, 5$. The toroidal inputs retain their admissible, nondegenerate parameter domains; the unrestricted wording on source pages 12–13 cannot include the unit period $(1, 0)$. Indeed its displayed rotation presentation gives $y = x^2$ and the cyclic group C_3, with coincident parabolics and no thin geometry. The exceptional $(1, 1)$ system is Example 4.4. We do not assert a larger parameter domain by extrapolating the finite tables. The positive rows above are verified independently of classification completeness.

Earlier attributed remarks about polytope dualities, the 11-cell, Schulte–Weiss rotations, chiral extensions, codimension-two faces and the Coxeter honeycomb construction remain contextual antecedents. They are not new classifications, additional finite table rows or hypotheses secretly used in Theorem 1. The $p = 6$ programme is explicitly deferred by this source; this synthesis does not turn that announcement into a new L052 obligation or a fresh intake.

4 Questions, evidence and attribution

Problems 9.1 and 9.2 and the unnumbered completeness conjecture for the known chiral list remain questions. Checking the positive rows does not prove the latter two classifications. The programme's earlier negative answers to Problems 9.3 and 9.4 are acquired with their precise standard parabolic hypotheses: in 9.3 thinness and chamber connectedness force the designated chamber orbit to be all chambers; in 9.4 the rank-three standard construction is residually connected. The latter uses a connected spanning coset graph, not an unjustified equality of incidence relations. These old answers receive no new question credit and retain their previous limited formalization scope.

The mathematical results and classical inputs keep the attributions in [1]. The general component was authored by the root collaborator and independently reviewed by the present assembler. The finite component was authored by a separate collaborator and independently reviewed by root. The present contribution is the complete source join and Proposition 2; the finite author confirmed its convention without supplying a new finite computation. The accepted chamber and L003 shared proofs remain complete acquired dependencies. Named group constructors, exact presentation/isomorphism routines and the named CHEVIE model are explicit software and identification premises of the finite certificate. No new Lean theorem, measurement, compression factor or priority claim is made. Whole-source admission still requires a distinct review of this assembly and the publication gate.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Math. 90 (2016), 1045–1067; fixed accepted author manuscript, 22 pages.

[2] *Residue stabilizers and distinguished generators*, accepted complete general recovery, eight pages; exact author, independent review and root-adoption editions acquired.

[3] *Finite coset geometries in Highly symmetric hypertopes*, accepted complete finite recovery, five pages; exact author and distinct root-review editions acquired.

[4] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002. Marked correspondence and the cited regular locally toroidal classifications retained.

[5] A. Breda D'Azevedo and R. Nedela, *Chiral hypermaps of small genus*, Beitr. Algebra Geom. 44 (2003), 127–143. Retained source input.

[6] D. Corn and D. Singerman, *Regular Hypermaps*, J. Combinatorics 9 (1988), 337–351, as cited in the fixed source.